

SymPy Bug Card

Bug 42: `solve_univariate_inequality(atan(x) < pi)` omits $x = 0$

Task. Solve the real univariate inequality

$$\arctan(x) < \pi.$$

SymPy's answer.

$$\text{solve_univariate_inequality}(\arctan(x) < \pi, x) = (-\infty, 0) \cup (0, \infty).$$

Correct answer.

$$\mathbb{R}.$$

Explanation. The returned set excludes 0, but $\arctan(0) = 0 < \pi$. More generally, real principal arctangent values lie in $(-\pi/2, \pi/2)$, so the inequality is true for every real x .

Likely root cause. The diagnosis identifies a branch-invalid inversion in `solveset`: $\tan(\pi) = 0$ is accepted as a root of $\arctan(x) = \pi$, and the inequality solver then removes this spurious root as a strict boundary. This is a new failure mode with no public precedent.

Artifact (GitHub).

[bug-report/bug-042-solve-univariate-inequality-atan-x-pi-omits-x-0/](https://github.com/sympy/sympy/issues/14442)